

Annotated Bibliography for Benchmark 2

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Ackerman, J. M., & Cybenko, G. (2021). Formal languages, deep learning, topology and algebraic word problems. Proceedings of the IEEE Security and Privacy Workshops (SPW 2021).

Summary.

Ackerman and Cybenko show that formal languages carry algebraic and topological structure, and that this structure can be used as a “signature” of the language. Using the Dyck-2 language as a case study, they demonstrate that its topology is equivalent to a figure-eight with two loops, a fixed invariant of the grammar. They then train neural networks on strings from Dyck-2 and use persistent homology to track how the topology of the internal representations changes across layers. The key finding is that the network systematically collapses the figure-eight topology into a simple, convex point cloud so that the classes become linearly separable, destroying the original topological invariants in the process.

Relevance.

This paper explains, at a mathematical level, why neural networks cannot preserve the formal guarantees that automata have: the representational regime itself is incompatible with the algebraic structure of formal languages. For our project, it anchors the representational loss theme—showing that when a parser’s internal representation ceases to encode a grammar’s invariants, properties like decidability, completeness, and length-independent generalization no longer have anything to “hook onto.” It justifies our claim that moving from automata to neural LMs is not a mere engineering trade-off but a change in what can be proven about the system.

Kamble, U., Salunke, Y., Nalawade, S., Dedgaonkar, S., Mahadik, T., & Shelke, P. (2024). Survey of application of automata theory in natural language processing. IEEE Conference Proceedings (ICETSI 2024).

Summary.

Kamble and colleagues survey how classical automata—finite-state automata, pushdown automata, and Turing machines—have been used across NLP tasks. They map FSAs to tokenization, morphological analysis, and pattern matching; PDAs to syntactic parsing and grammar checking; and Turing-complete models to semantic interpretation and verification tasks. For each level, they emphasize provable properties such as determinism, soundness, completeness, and decidability. They also discuss the practical limitations of automata-based systems on large, noisy corpora and describe emerging hybrid architectures that combine automata with neural components to balance scalability and formal guarantees.

Relevance.

This survey provides the baseline guarantees we are comparing against: it tells us exactly which properties automata-based parsers can prove and where they are used in real NLP systems. It also documents the real-world pressure to replace these components with neural models for scalability. In our project, we use Kamble et al.'s taxonomy to define correctness, completeness, and verifiability in concrete task terms and to motivate our stress-test designs (e.g., FSAs for regular patterns, PDAs/CFGs for context-free patterns).

Ackerman, J. M., & Cybenko, G. (2020). A survey of neural networks and formal languages. arXiv preprint arXiv:2006.01338.

Summary.

This survey analyzes how different neural architectures relate to classical formal language classes under input-bounded, finite-precision (IBFP) assumptions. The authors show that CNNs with max pooling recognize strictly local languages, weaker than full regular languages; SRNNs and GRUs are effectively bounded to regular languages; LSTMs can simulate simplified k-counter machines and thus recognize some counter languages, but not all context-free languages; and Transformers with periodic positional encodings are asymptotically weaker than full regular recognition. They also discuss memory-augmented networks (e.g., differentiable stacks), which can approach context-sensitive power but at the cost of training instability and architectural complexity.

Relevance.

This paper is the backbone of our structural loss theme. It lets us transform the vague claim “neural models lack formal guarantees” into the precise statement “each major architecture collapses to a particular, weaker language class, so guarantees for stronger classes are impossible.” We use its results to select test languages—Dyck-style parenthesis languages, center-embedding, and simple

a

n

b

n

a

n

b

n

analogues—that sit at or above the neural architectures’ proven limits, and to argue that PDA-level guarantees simply cannot be replicated by standard LSTMs or Transformers.

Oh, B.-D., Clark, C., & Schuler, W. (2022). Comparison of structural parsers and neural language models as surprisal estimators. *Frontiers in Artificial Intelligence*, 5, Article 777963.

Summary.

Oh and colleagues compare a left-corner parser based on a probabilistic context-free grammar with neural language models such as GPT-2, using surprisal as the evaluation metric on reading-time data. The structural parser assigns probabilities via explicit parse trees and grammar rules, whereas GPT-2 uses learned attention over embeddings with no explicit syntax. Empirically, the parser matches or outperforms GPT-2 in predicting self-paced reading times and eye-tracking measures, and larger GPT-2 models sometimes fit the human data worse. Importantly, the parser’s failures are systematic and traceable to missing or mis-weighted grammar rules, while the neural model’s errors are scattered and hard to interpret.

Relevance.

This study is central to our verifiability theme. It shows that even when a neural LM matches a structural parser on a behavioral metric (surprisal), it does not gain the parser's formal guarantees: only the parser is provably sound and complete with respect to its CFG, and its error profile is diagnosable. We use this as direct support for our claim that empirical performance parity does not equal theoretical equivalence, and as motivation for comparing our own FSA/CFG models with a probabilistic language model on controlled syntactic stimuli.

Miaschi, A., Brunato, D., Dell'Orletta, F., & Venturi, G. (2022). Robustness and sensitivity of a neural language model: A case study on Italian L1 learner errors. IEEE Access.

Summary.

Miaschi et al. evaluate BERT on Italian learner essays that contain grammatical, orthographic, and lexical errors. They design two probing settings: one measures semantic robustness by testing whether sentence similarity judgments remain stable under injected errors, and the other measures linguistic sensitivity by training classifiers on BERT's layer representations to predict morphosyntactic features. They find that certain error types—especially orthographic errors—cause sharp drops in semantic similarity and degrade the classifiability of syntactic information, with effects varying across layers. The degradation patterns are complex and cannot be captured by a simple rule system.

Relevance.

This paper provides empirical evidence for our behavioral loss theme. It shows that, in the absence of formal guarantees, a neural LM's behavior under noisy or ungrammatical input is neither predictable nor easily interpretable. In contrast, an automaton-based parser would respond deterministically to the same inputs, accepting or rejecting according to explicit rules and offering a trace of where the error occurs. We draw on these findings to justify the error-injection aspects of our own stress-test and to argue that unpredictability under grammatical uncertainty is a core consequence of replacing formal grammars